# Unifying Theory of Dynamic Aether Mechanics: Special Relativity

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**Abstract**

Within the Dynamic Aether Mechanics framework, special relativity emerges naturally from the properties of a compressible quantum superfluid medium. This paper derives the Lorentz factor $\gamma = 1/\sqrt{1 - v^2/c^2}$ from geometric path-length effects in the medium, time dilation from both velocity (path geometry) and gravity (inflow-velocity-dependent time dilation), and mass–energy equivalence $E = mc^2$ from the compression wave energy of dissolving mass. The five classical experiments historically regarded as having disproved aether theories—Michelson–Morley, Kennedy–Thorndike, Sagnac, stellar aberration, and Fizeau—are all quantitatively predicted by the model. The linearity of the medium's pressure–density response is confirmed to $10^{-8}$ precision across velocities from $10^{-6}\,c$ to $0.9994c$.

*This paper is part of a series presenting the Unifying Theory of Dynamic Aether Mechanics. For the foundational concepts of Aether binding, the three independent properties (density, pressure, temperature), and the unification of force constants, see Paper 1: General Overview [12].*

## 1. Special Relativity

The greatest challenge to this theory is not internal failures, but the extraordinary success of special relativity. However, many effects predicted by special relativity also emerge naturally from dynamic Aether.

### 1.1 Length Contraction

Every force that binds matter—electromagnetic, strong, weak—propagates through the Aether at the local speed of sound $c = \sqrt{K_A/\rho_A}$. The wave equation governing this propagation, $\partial^2\phi/\partial t^2 = c^2\,\nabla^2\phi$, is invariant under the Lorentz transformation [6]. This invariance guarantees that if a bound system is in equilibrium at rest, the transformed configuration—contracted by $1/\gamma$ in the direction of motion, with perpendicular dimensions unchanged—is also a valid equilibrium for the same system moving at $v$ through the medium. Lorentz [6] proved this rigorously for electromagnetic forces; in this theory the result extends to all matter, because every binding force propagates through the Aether at $c$. Length contraction is therefore not an *ad hoc* hypothesis but a mathematical consequence of all forces sharing the same medium and the same propagation speed.

The anisotropy that drives the contraction is visible in the force propagation times of a moving system. Force signals traveling along the direction of motion encounter asymmetric propagation: upstream at effective speed $c - v$, downstream at $c + v$. The round-trip signal time along a separation $L$ parallel to the motion is

$$t_\parallel = \frac{L}{c-v} + \frac{L}{c+v} = \frac{2L/c}{1 - v^2/c^2} \tag{1}$$

while the round-trip time across the same separation perpendicular to the motion is

$$t_\perp = \frac{2L}{\sqrt{c^2 - v^2}} = \frac{2L/c}{\sqrt{1 - v^2/c^2}} \tag{2}$$

These differ by a factor of $\gamma$: force signals along the direction of motion are delayed relative to perpendicular signals. In the contracted equilibrium ($L' = L/\gamma$ parallel, $L$ perpendicular), this anisotropy is absorbed into the steady-state field configuration—the retarded fields of all the moving constituents self-consistently produce the same binding as in the rest frame.

Note that this contraction is experimentally confirmed only for *velocity-driven* motion through the Aether. In the gravitational case, the Aether's natural coordinate system (Painlevé–Gullstrand coordinates) has flat spatial slices: the inflow produces time dilation (Section 1.3) but no radial spatial distortion. Gravitational length contraction has no experimental confirmation.

A secondary consequence of velocity contraction is compression of the Aether in the rest frame of the moving object. Because the object does real work on the medium, a genuine density increase $\delta\rho/\rho_A \approx v^2/(2c^2)$ develops via Bernoulli (see Paper 7 [13], Section 6.2 for the vortex-level treatment). This compression is a consequence of the contraction, not its cause—the cause is the Lorentz invariance of the wave equation. The compression is physically distinct from gravitational inflow, where free-fall kinetic energy exactly balances potential energy, leaving $\Delta\rho = 0$.

### 1.2 Time Dilation—Velocity

All internal processes of an object—atomic oscillations, wave propagation within its structure, binding cycles—depend on waves that propagate at speed $c$ relative to the Aether, not relative to the object. When the object moves at velocity $v$ through the Aether, these internal waves must travel longer geometric paths through the medium.

Consider any wave-based process within the moving object. A wave crossing the object perpendicular to its direction of motion covers distance $ct$ in the Aether frame while the object moves $vt$. By the Pythagorean theorem, the perpendicular distance covered by the wave is only $\sqrt{c^2 - v^2}\,t$. The internal process that took time $t_0$ at rest now takes time $t_0/\sqrt{1-v^2/c^2}$—exactly the Lorentz factor $\gamma$.

Since every force in this theory (electric, magnetic, gravitational) propagates through Aether at speed $c$, every internal process of the moving object is slowed by the same factor. Time is fundamentally a measure of relative particle movement (a year measures Earth's orbital motion; a clock measures the motion of its internal mechanisms), so this universal slowing produces the observed time dilation.

Note that time dilation and length contraction share the same root cause—motion through a medium with finite propagation speed $c$—and both follow from the Lorentz invariance of the wave equation governing that medium. Time dilation arises from longer *wave paths* (Pythagorean theorem); length contraction arises from the *anisotropic retarded-field configuration* of a moving bound system, whose equilibrium contracts by $1/\gamma$ in the direction of motion [6].

The factor $v^2/c^2$ that governs time dilation is the same factor that appears in the magnetic-to-electric force ratio ($F_{\text{mag}} = F_{\text{elec}} \cdot v_1 v_2/c^2$) and in relativistic energy ($E = \gamma m c^2$). All four arise from the same Aether mechanics: the ratio of motion velocity to medium propagation speed. This connection is developed in the Unification of Force Constants section.

### 1.3 Time Dilation—Gravitational

Near massive objects, the Aether flows inward—drawn by the collective electron vortices of the mass through the transient bind–unbind cycle described in Section 1. This inflow is the direct cause of gravitational time dilation. A static observer—one who maintains a fixed position while the Aether flows past—has all internal processes affected by the relative motion between their wave-based structure and the medium. Because internal processes depend on waves propagating at the local speed $c = \sqrt{K_A/\rho_A}$ *relative to the Aether*, a static observer in a flowing medium experiences a slowing of all processes identical to velocity time dilation.

The inflow velocity at distance $r$ from a mass $M$ follows from energy conservation (an Aether element falling from rest at infinity):

$$v_{\text{in}}(r) = \sqrt{\frac{2GM}{r}} \tag{3}$$

A static observer at radius $r$ is stationary while the Aether flows past at $v_{\text{in}}$. The time dilation experienced by this observer is:[1]

$$\frac{\Delta t}{t} = \frac{v_{\text{in}}^2}{2\,c^2} = \frac{GM}{r\,c^2} \tag{4}$$

[1] A density-based approach would require $\Delta\rho/\rho_A = 2GM/(rc^2)$ to reproduce the observed time dilation from $c$-variation alone. However, Bernoulli's equation for the Aether's logarithmic equation of state shows this is unphysical: with inflow at $v_{\text{in}} = \sqrt{2GM/r}$, the kinetic energy exactly balances the gravitational potential energy, leaving zero energy for compression ($\Delta\rho/\rho_A = 0$ exactly). The maximum density perturbation (hydrostatic equilibrium, $v = 0$) is $GM/(rc^2)$—half the required value. The flow-based derivation resolves this by attributing time dilation to the inflow velocity rather than to a density change.

This matches the observed gravitational time dilation confirmed by the Pound–Rebka experiment, the Gravity Probe A mission, and the GPS satellite system.

The Aether density $\rho_A$ and the speed of light $c = \sqrt{K_A/\rho_A}$ remain constant throughout the inflow. This follows from Bernoulli's equation for the Aether's logarithmic equation of state ($P = K_A \ln(\rho/\rho_0)$): along a streamline, $\frac{1}{2}v_{\text{in}}^2 + h(\rho) + \Phi = \text{const}$. For free-fall inflow ($\frac{1}{2}v_{\text{in}}^2 = GM/r$ and $\Phi = -GM/r$, so $\frac{1}{2}v_{\text{in}}^2 + \Phi = 0$), the enthalpy terms cancel exactly, giving $\rho = \rho_0$ everywhere. This result is equation-of-state independent: for *any* barotropic fluid in free-fall, the kinetic energy equals the gravitational potential energy, leaving no energy for compression.

Both velocity and gravitational time dilation now arise from a single mechanism: motion relative to the local Aether. Velocity time dilation occurs when an object moves through the Aether; gravitational time dilation occurs when the Aether moves past a static object. In both cases, the fractional time dilation is $v_{\text{rel}}^2/(2c^2)$, where $v_{\text{rel}}$ is the velocity between the object and the local Aether. A freely falling observer—one who co-moves with the inflowing Aether—experiences $v_{\text{rel}} = 0$ and therefore *no* time dilation, recovering the equivalence principle from the medium's dynamics. $G$ connects directly to the medium: it measures the strength of the inflow driven by the bind–unbind cycle, expressed in units of the medium's wave speed ($c^2 = K_A/\rho_A$).

### 1.4 The Event Horizon and Inflow Conservation

The inflow velocity $v_{\text{in}}(r) = \sqrt{2GM/r}$ (Eq. 3) is a steady-state *circulation*, not a one-way accumulation. Aether flows inward, participates in transient binding near the mass, and is returned upon pair annihilation—only the tiny asymmetry fraction $\epsilon$ escapes as gravitational radiation (see Section 2.1.4 for the detailed conservation analysis).

This inflow has two further consequences:

1. **Gravitational lensing.** Light propagating through the inflowing Aether follows geodesics of the acoustic metric—the effective spacetime geometry experienced by waves in a moving medium. For an inflow at $v_{\text{in}} = \sqrt{2GM/r}$ with uniform wave speed $c$, the acoustic metric takes the Painlevé–Gullstrand form, which is exactly equivalent to the Schwarzschild geometry. The total deflection angle is $4GM/(r_0\, c^2)$, matching general relativity (Section 2.1).

2. **The event horizon.** At the Schwarzschild radius $r_s = 2GM/c^2$, the inflow velocity equals $c$. Outward-propagating waves can no longer escape, producing a sonic horizon analogous to those in supersonic fluid dynamics. Inside this radius, the transient binding cycle still operates locally, but the restoring pressure pulses from pair annihilation are swept inward by the superluminal flow.

### 1.5 Mass–Energy Equivalence ($E = mc^2$)

This theory requires that Aether can bind into mass. It follows that mass can also unbind—dissolving back into Aether. The energy released by this dissolution can be calculated, and the result connects directly to the transient binding mechanism that drives gravity. In quantum field theory terms, this dissolution is annihilation—the same process that ends each virtual particle cycle (and the reverse of the permanent binding produced by the Dynamic Casimir Effect).

When a mass dissolves into Aether, its volume before dissolution is negligible compared to its volume as unbound Aether (we are considering the volume of the actual particle matter, not the empty space within atoms). The dissolved Aether must displace the surrounding ambient Aether, and the total mass of Aether displaced equals the original mass (imagine a box perfectly filled with identical marbles—forcing one additional marble in displaces exactly one marble's worth of mass).

This displacement creates a compression wave propagating at the speed of light ($c$), since light speed is the propagation speed of compression waves in Aether.

The total energy carried by the compression wave follows from the bulk modulus $K_A$ and the displaced volume $V = m/\rho_A$:

- **Elastic energy** from the compression: the energy stored per unit volume displaced in a medium of bulk modulus $K_A$ is $\frac{1}{2}K_A$ (at unit strain). For volume $V = m/\rho_A$: $E_p = \frac{1}{2}K_A\,(m/\rho_A) = \frac{1}{2}(K_A/\rho_A)\,m = \frac{1}{2}mc^2$.
- **Kinetic energy**: for a progressive compression wave in a linear medium, the time-averaged kinetic and elastic energies are equal (equipartition), so $E_k = E_p = \frac{1}{2}mc^2$.
- **Total wave energy**: $E = E_k + E_p = mc^2$.

The result depends only on $c^2 = K_A/\rho_A$: the energy per unit mass dissolved equals the square of the wave speed. This is the standard energy–mass relation for compression waves in an elastic medium.

In the Aether model, this equation represents the electromagnetic energy released when mass of negligible pre-dissolution volume dissipates into ambient Aether—and equivalently, the energy required to compress Aether into mass (the energy cost paid when the DCE threshold is exceeded and a binding event becomes permanent). Each transient binding cycle in the gravity mechanism momentarily invests $mc^2$ of energy to form a virtual pair and recovers it upon annihilation. The gravitational wave carries only the tiny asymmetry of this process outward. The concept of mass as condensed energy, central to relativity, is independently predicted by compressible Aether.

### 1.6 Relativistic Kinetic Energy

The preceding sections derived two results purely from Aether mechanics: (1) all internal processes of an object moving at velocity $v$ through the medium slow by a factor $\gamma = 1/\sqrt{1 - v^2/c^2}$, from the geometric path-length increase of internal waves; and (2) $E = mc^2$, the energy stored in a mass at rest. These two results together determine the kinetic energy of a moving object without invoking any further assumptions.

Consider a particle (a vortex excitation) being accelerated from rest to velocity $v$ by an applied force $F$. The particle's structural integrity is maintained by its internal circulation—the same wave processes that experience time dilation. As the particle moves faster, these internal processes run slower by the factor $\gamma$, which means the particle's inertial response to the applied force decreases: its internal dynamics can redistribute momentum only at the dilated rate. The effective momentum of the moving particle is therefore:

$$p = \gamma m v \tag{5}$$

This is not an additional postulate—it follows directly from the time dilation already derived. A vortex whose internal clock runs $\gamma$ times slower responds to force $\gamma$ times more sluggishly, producing an effective inertia of $\gamma m$.

The kinetic energy is the work done to accelerate the particle from rest, $W = \int_0^v v' \, dp$. Evaluating with $p = \gamma m v$ gives:

$$\mathrm{KE} = (\gamma - 1)\, mc^2 \tag{6}$$

At low velocities, $\gamma \approx 1 + v^2/(2c^2)$, recovering the classical kinetic energy $\frac{1}{2}mv^2$. As $v \to c$, the energy diverges: the compression waves that constitute the object's internal structure can no longer propagate ahead of the object, and infinite energy would be required to reach the medium's wave speed.

The total energy of a moving particle is thus $E_{\text{total}} = \gamma mc^2$: rest energy plus kinetic energy, both arising from the mechanics of a vortex excitation in a compressible medium with finite propagation speed $c$. This completes the triad of $v^2/c^2$ effects—time dilation, magnetic force, and relativistic energy—all derived from the same Aether physics without recourse to special-relativistic postulates.

## 2. Classical Aether Experiments

The experiments of the 19th and early 20th centuries are widely regarded as having disproved the existence of Aether. However, they disproved only the rigid, static luminiferous aether of that era—a medium assumed to have no physical effect on the objects moving through it. Dynamic Aether—a medium in which all forces propagate at speed $c$, producing length contraction, time dilation, and inflow-dependent gravitational effects—predicts the results of every classical aether experiment.

Five key experiments and their explanations follow.

### 2.1 The Michelson–Morley Experiment

The Michelson–Morley experiment [7] (1887) measured the round-trip travel time of light along two perpendicular arms of an interferometer. If Earth moves through a static Aether at velocity $v$, light traveling along the arm parallel to Earth's motion should take longer than light traveling perpendicular to it, producing an interference fringe shift.

In this theory, the null result is predicted by length contraction. An object moving at velocity $v$ through the Aether is contracted in the direction of motion by the factor $\sqrt{1 - v^2/c^2}$, because the binding forces holding the interferometer together propagate at $c$ through the medium and their retarded-field equilibrium contracts by this factor (Section 1). The parallel arm of the interferometer is shortened by exactly $1/\gamma$, which precisely compensates for the longer light travel time along that arm.

The round-trip travel time for light along the parallel arm is $t_{\|} = (2L/c)/(1 - v^2/c^2)$, while the perpendicular round-trip is $t_{\perp} = (2L/c)/\sqrt{1 - v^2/c^2}$ (Section 1). These differ by a factor of $\gamma$. With the parallel arm contracted to $L' = L/\gamma$, the contracted parallel time becomes $t_{\|}(L') = (2L'/c)/(1 - v^2/c^2) = (2L/c)/\sqrt{1 - v^2/c^2} = t_{\perp}$. The fringe shift is exactly zero—not because the Aether does not exist, but because it contracts the apparatus in a way that conceals its own presence.

### 2.2 The Kennedy–Thorndike Experiment

The Kennedy–Thorndike experiment [8] (1932) modified the Michelson–Morley design by using arms of deliberately different lengths. Length contraction alone cannot produce a null result when the arms are unequal, because the contraction factor cancels only when both arms are the same length. A non-null result would therefore reveal that length contraction is the sole effect of motion through the medium.

The null result of this experiment is explained by this theory because it predicts both length contraction and time dilation. The combination of spatial contraction (shortening the parallel arm) and temporal dilation (slowing the clock that measures the fringe pattern) together produce a complete null result for any arm length ratio. This is the same combination that special relativity invokes, but here both effects arise from the physical mechanics of moving through a medium in which all forces propagate at finite speed $c$.

### 2.3 The Sagnac Effect

The Sagnac effect [9] (1913) is the observed fringe shift when light travels around a rotating loop—light traveling with the rotation arrives later than light traveling against it. Unlike the Michelson–Morley null result, the Sagnac effect is a positive detection of motion.

This result is more naturally explained by a medium theory than by special relativity. If light propagates at speed $c$ relative to the Aether (not relative to the apparatus), then in a rotating interferometer the co-rotating beam must travel a longer path (the far mirror has moved away) while the counter-rotating beam travels a shorter path (the near mirror has moved closer). The path difference produces a fringe shift proportional to the angular velocity and the enclosed area:

$$\Delta t = \frac{4A\omega}{c^2} \tag{7}$$

where $A$ is the enclosed area and $\omega$ is the angular velocity. This is exactly the observed result. In the Aether model, the explanation is straightforward: light moves at $c$ relative to the medium, and the rotating apparatus moves relative to both. The Sagnac effect is direct evidence that light speed is determined by a local reference—precisely what a medium theory predicts.

Modern ring laser gyroscopes and fiber optic gyroscopes exploit this effect for navigation, confirming the Sagnac fringe shift to extraordinary precision.

### 2.4 Stellar Aberration

Stellar aberration—the apparent displacement of star positions due to Earth's orbital velocity—was first observed by Bradley in 1728 [10]. A telescope must be tilted slightly in the direction of Earth's motion to center a star's image, by an angle of approximately 20.5 arc-seconds.

In this theory, the explanation is direct: Earth moves through the Aether at orbital velocity $v$, while starlight propagates at $c$ through the same medium. The telescope must be tilted by angle $\theta$ where $\tan\theta = v/c$ to compensate for Earth's motion during the time light traverses the telescope tube.

This result historically posed a dilemma for 19th-century aether theories. If the Aether were dragged along with Earth, aberration should not occur (the medium and telescope would move together). If the Aether were stationary, the Michelson–Morley experiment should have detected Earth's motion. Dynamic Aether resolves this dilemma: Earth moves freely through the Aether (producing aberration), while length contraction and time dilation (physical consequences of that motion) eliminate the Michelson–Morley signal. The two results, long considered contradictory for any aether model, are simultaneously explained.

### 2.5 The Fizeau Experiment and the Fresnel Drag Coefficient

The Fizeau experiment [11] (1851) measured the speed of light in moving water. If water fully drags the Aether with it, light in the moving water should travel at $c/n + v$ (where $n$ is the refractive index and $v$ is the water velocity). If water does not drag the Aether at all, light should travel at $c/n$ regardless of water motion.

Fizeau found neither result. Instead, the speed of light in moving water is:

$$c_{\text{observed}} = \frac{c}{n} + v\left(1 - \frac{1}{n^2}\right) \tag{8}$$

The factor $(1 - 1/n^2)$ is the Fresnel drag coefficient [4]—partial dragging, where only a fraction of the medium's velocity adds to the light speed.

This result derives naturally from the Aether framework. A material with refractive index $n$ slows light because it increases the effective Aether density within the material's volume. (This is distinct from gravitational fields, where Bernoulli gives $\Delta\rho_A = 0$.) Since $c = \sqrt{K_A/\rho_A}$ and the material reduces light speed by factor $n$, the effective density within the material is:

$$\rho_{\text{material}} = n^2\,\rho_A \tag{9}$$

The excess density above the ambient background is:

$$\Delta\rho = (n^2 - 1)\,\rho_A \tag{10}$$

When the material moves at velocity $v$, only this excess density moves with it. The ambient Aether density $\rho_A$ remains stationary—it is the background medium. The fraction of the total density that is dragged with the material is:

$$f_{\text{drag}} = \frac{\Delta\rho}{\rho_{\text{material}}} = \frac{(n^2 - 1)\,\rho_A}{n^2\,\rho_A} = \frac{n^2 - 1}{n^2} = 1 - \frac{1}{n^2} \tag{11}$$

This is exactly the Fresnel drag coefficient. The partial dragging is not an empirical correction but a direct consequence of the density structure of Aether within refractive media. The ambient Aether is stationary; only the material's contribution to the local density moves with the material.

This derivation also explains why the drag coefficient depends only on the refractive index: it is purely a density ratio, independent of the material's chemical composition, temperature (except through $n$), or other properties.

This derivation carries a significant implication for the bulk modulus $K_A$. The relationship $\rho_{\text{material}} = n^2\rho_A$ was derived by assuming that $K_A$ remains constant inside the material—that the material changes only the density, not the stiffness, of the Aether. If the material also altered $K_A$, then

$c_{\text{material}} = \sqrt{K_{\text{material}}/\rho_{\text{material}}}$ and the Fresnel drag coefficient would depend on both the density ratio and the stiffness ratio, not solely on the refractive index. The experimental success of the $1 - 1/n^2$ coefficient confirms that $K_A$ is indeed invariant inside materials—the same conclusion reached independently from velocity time dilation, but here confirmed across a vastly wider range of density variation.

The density relationship also means that every refractive index measurement is a measurement of local Aether density. Water ($n = 1.33$) multiplies the ambient Aether density by 1.77. Glass ($n = 1.52$) multiplies it by 2.31. Diamond ($n = 2.42$) multiplies it by 5.86. The entire field of optics becomes, in this framework, a body of evidence about how different materials modify the local Aether density—with refractive index serving as the density probe.

## 3. Summary of Theoretical Properties

### 3.1 Properties of Interferometric Predictions

- The Michelson–Morley null result is predicted by length contraction: the parallel arm contracts by $\sqrt{1 - v^2/c^2}$, exactly compensating for the directional light travel time difference.
- The Kennedy–Thorndike null result requires both length contraction and time dilation, both of which this theory independently predicts from motion through the medium.
- The Sagnac effect is a direct prediction of light propagating at $c$ relative to the medium: rotating apparatus creates an asymmetric path length, producing a fringe shift of $\Delta t = 4A\omega/c^2$.
- Stellar aberration is explained by Earth's motion through Aether at orbital velocity, requiring telescope tilt of $v/c$—approximately 20.5 arc-seconds.
- The Fresnel drag coefficient $(1 - 1/n^2)$ derives from the density structure of Aether in refractive media: material with refractive index $n$ creates effective density $n^2\rho_A$, of which only the excess $(n^2 - 1)\rho_A$ moves with the material while ambient $\rho_A$ remains stationary.
- The Fresnel drag derivation independently confirms that $K_A$ (bulk modulus) is invariant inside materials: if $K_A$ changed with density, the drag coefficient would deviate from $1 - 1/n^2$. This extends the linearity confirmation from density perturbations of order $v^2/c^2$ to density ratios up to 6:1.
- Every refractive index measurement is a measurement of local Aether density: a material with refractive index $n$ multiplies the ambient density by $n^2$.
- The gravitational lensing bending angle $4GM/(r_0c^2)$ arises entirely from the acoustic metric of the inflowing Aether: transverse advection by the flow $(2GM/(r_0c^2))$ and time-delay

asymmetry from the flow ($2GM/(r_0 c^2)$), matching general relativity without spacetime curvature.

- The classical aether experiments, once considered fatal to aether theories, are fully explained by a medium in which all forces propagate at $c$, producing length contraction and time dilation as physical consequences of motion.

## 4. Experimental Support and Connections to Established Physics

The following experimentally confirmed phenomena support or align with the aspects of the theory presented in this paper:

1. **Relativistic E/B field mixing.** In special relativity, a pure electric field in one reference frame appears as a combination of electric and magnetic fields in a frame moving relative to the first. This is a direct consequence of the Aether model: a stationary charge has a purely radial flow vector (**E** only), but in a frame where the charge moves, that flow vector is tilted by the motion, developing angular structure (**B** appears).

2. **Precision velocity time dilation.** The Lorentz factor $\gamma = 1/\sqrt{1 - v^2/c^2}$ has been confirmed across velocities from $10^{-6}\,c$ (GPS [2], Mössbauer [3]) to $0.9994c$ (CERN muon storage ring), with no higher-order corrections ($v^4/c^4$ terms) detected to $10^{-8}$ precision. In the Aether model, this confirms that the medium is perfectly linear in its pressure–density response. The Weber–Kohlrausch experiment [1] independently confirmed that the $c$ in the electromagnetic force ratio is the same $c$ as the speed of light.

3. **The Michelson–Morley null result.** The Michelson–Morley experiment [7] found no directional dependence of light speed despite Earth's orbital velocity through space. This theory predicts exactly this result: length contraction of the parallel interferometer arm by the factor $\sqrt{1 - v^2/c^2}$ precisely compensates for the upstream/downstream travel time difference, producing zero fringe shift.

4. **The Kennedy–Thorndike null result.** The Kennedy–Thorndike experiment [8] used unequal arm lengths, where length contraction alone cannot produce a null result. This theory predicts the null result because it provides both length contraction and time dilation—the same combination that special relativity requires, but derived from physical motion through a medium in which all forces propagate at $c$.

5. **The Sagnac effect.** Observed in 1913 [9] and confirmed to high precision in modern ring laser gyroscopes, the fringe shift in a rotating interferometer ($\Delta t = 4A\omega/c^2$) is a direct prediction of light propagating at $c$ relative to the medium rather than relative to the apparatus.

6. **Stellar aberration.** Bradley's 1728 observation [10] of 20.5 arc-second stellar displacement due to Earth's orbital velocity is directly predicted by Earth's motion through Aether. This theory simultaneously explains why aberration occurs (Earth moves through the medium) and why the Michelson–Morley experiment is null (length contraction and time dilation cancel the signal), resolving a long-standing dilemma for aether models.

7. **The Fresnel drag coefficient.** Fizeau's 1851 measurement [11] of light speed in moving water found partial dragging by the factor $(1 - 1/n^2)$. This coefficient derives directly from the Aether density framework: material with refractive index $n$ creates effective density $n^2\rho_A$, of which only the excess $(n^2 - 1)\rho_A$ moves with the material.

8. **$K_A$ invariance across refractive media.** The Fresnel drag coefficient $(1 - 1/n^2)$ is exact only if the bulk modulus $K_A$ remains constant inside materials—if $K_A$ varied with density, the coefficient would acquire additional terms. The experimental exactness of $1 - 1/n^2$ across materials with widely differing refractive indices confirms $K_A$ invariance at density ratios up to $n^2 \approx 6$ (for diamond), extending the linearity confirmation by three orders of magnitude beyond the velocity-based experiments.

## 5. Open Questions and Testable Predictions

The following areas relevant to this paper remain underdeveloped and represent opportunities for further refinement or falsification:

1. **Extending the Fizeau framework to dispersive media.** The derivation of the Fresnel drag coefficient from $\rho_{\text{material}} = n^2\rho_A$ assumes a single refractive index. In dispersive media, $n$ varies with wavelength, and Lorentz [5] showed that the drag coefficient acquires a dispersion correction: $f = 1 - 1/n^2 - (\lambda/n)(dn/d\lambda)$. Deriving this correction from the Aether density model would strengthen the connection between refractive index and Aether density.

## 6. Mathematical Summary

This section collects the key equations relevant to the topics covered in this paper. *For the complete mathematical summary, see Paper 1: General Overview [12].*

### 6.1 Fundamental Aether Parameters

The theory posits four microscopic parameters from which all macroscopic phenomena derive:

| Symbol | Name | Constrained range | Primary constraint |
|---|---|---|---|
| $\rho_A$ | Ambient density | $10^{11}$–$10^{12}$ kg/m$^3$ | Electron vortex energy |
| $K_A$ | Bulk modulus | $10^{28}$–$10^{29}$ J/m$^3$ | $K_A = \rho_A c^2$ |
| $m_A$ | Aether quantum mass | 500–850 MeV/$c^2$ | Flux tube width / glueball mass |
| $\xi$ | Healing length | 0.17–0.28 fm | Bridge diameter (lattice QCD) |

These four are not independent. Two defining relations connect them:

$$c = \sqrt{K_A/\rho_A} \tag{12}$$

$$\xi = \frac{\hbar}{m_A\, c\, \sqrt{2}} \tag{13}$$

leaving two free Aether parameters (e.g., $\rho_A$ and $m_A$) from which all others follow. (A third parameter, $\epsilon$, enters through gravity; see below.)

### 6.2 Equation of State

The Aether obeys a logarithmic equation of state, the unique form that produces a density-independent bulk modulus (Section 4):

$$P = K_A \ln\!\left(\frac{\rho}{\rho_0}\right) + P_0 \tag{14}$$

Key consequence: the speed of sound $c_s = \sqrt{K_A/\rho}$ varies with density. At ambient density, $c_s = c$. Note: in gravitational fields, Bernoulli's equation for free-fall inflow gives $\Delta\rho/\rho_A = 0$ exactly, so $c$ does not vary near gravitating masses; the density dependence is relevant for material media and for the two-fluid cosmological structure. This equation of state guarantees that $K_A$ is constant under compression, which is experimentally confirmed by:

- Velocity time dilation measurements spanning $10^{-6}\,c$ to $0.9994\,c$ (precision $\sim 10^{-8}$).
- The Fresnel drag coefficient $1 - 1/n^2$, exact across materials with density ratios up to ~6:1.

### 6.3 Special Relativity

All relativistic effects arise from motion through the medium.

**Lorentz factor.**

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}} \tag{15}$$

Derived geometrically from the ratio of helical to straight path lengths in the medium.

**Mass–energy equivalence.**

$$E = m\,c^2 \tag{16}$$

Mass is bound Aether; $c^2 = K_A/\rho_A$ is the medium's elastic energy per unit density. When mass dissolves, it releases a compression wave with this energy.

**Velocity time dilation.** A moving vortex must traverse a longer helical path through the medium. All internal processes slow by $\gamma$:

$$t_{\text{moving}} = \gamma\, t_{\text{rest}} \tag{17}$$

**Length contraction.** All binding forces propagate at $c$; the shifted retarded-force equilibrium contracts bound systems along the direction of motion:

$$L_{\text{moving}} = \frac{L_{\text{rest}}}{\gamma} \tag{18}$$

**Fresnel drag coefficient.**

$$f = 1 - \frac{1}{n^2} \tag{19}$$

where $n$ is the refractive index. A material with index $n$ has local Aether density $n^2\,\rho_A$; only the excess $(n^2 - 1)\rho_A$ moves with the material.

### 6.4 Linearity Constraint

The theory requires that all density perturbations from known particle interactions are small compared to $\rho_A$. The largest perturbation from the electron's own gravitational field occurs at its Compton wavelength, where the escape velocity is $v_{\text{esc}} = \sqrt{2Gm_e/\lambdabar_C}$ and the corresponding density ratio is $v_{\text{esc}}^2/c^2$:

$$\frac{\delta\rho}{\rho_A} = \frac{v_{\text{esc}}^2}{c^2} = \frac{2\,G\,m_e}{\lambdabar_C\,c^2} \approx 3.5 \times 10^{-45} \ll 1 \tag{20}$$

This extraordinary smallness confirms the Lorentz factor to 45 decimal places and guarantees that the linear approximation underlying special relativity is valid for all accessible energies. Deviations from exact Lorentz symmetry are predicted only at compressions approaching $\delta\rho/\rho_A \sim 1$, far beyond current experimental reach.

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